

SPECIFICATIONS

DELIVERY AND INSTALLATION OF NEW BALE PRESS, STEADY FLOW BREAKER, AUTOMATED CONVEYOR, BALE BAGGING AND HEAVY-DUTY BRACING SYSTEM

USDA, AMS, Cotton & Tobacco Program, Memphis, TN

C.1 SCOPE:

The United States Department of Agriculture (USDA), Agricultural Marketing Service (AMS), Cotton and Tobacco (C&T) Program in Memphis, TN requires a Contractor to provide a new full-eject bale press with a steady flow breaker, automated conveyor system with an bale bagging and heavy duty bracing system (see note within Section C.4) to be delivered and installed at the USDA, AMS, C&T Program Memphis Classing Office located at 3275 Appling Rd Memphis, TN 38133. The Contractor is also required to disassemble and remove the existing bale press and provide a *minimum* one (1) year warranty (see Sec. C.7) for the new bale press system.

The Contractor must furnish all labor, materials, equipment, parts, permits, insurance, supervision and transportation necessary to complete the removal of the existing system and the delivery and installation of the entirety of the new system full-eject bale press with a steady flow breaker, automated conveyor system with integrated scale, bale bagging and heavy duty bracing system to include the removal of the old bale press system (see Sec. C.5) within six (6) months of the date of the award.

C.2 DEFINITIONS:

Full Eject Bale Press - A bale press that compresses cotton into a specified dimension and then ejects the compressed and tied bale before accumulating more cotton for the next bale.

Steady Flow Breaker – A collections of rollers that evenly distribute cotton samples inside the baler infeed to guarantee consistent bale density.

Bale Bagging System - A automated system that wraps the bale in plastic to avoid contamination for certain materials, such as textile fibers, films, and nonwovens.

Heavy Duty Bracing System – A bracing system that can continuously support up to 500lbs that will prevent a bale from tipping over and damaging the wall and floor.

C.3 PROCESS DESCRIBED:

The cotton classing office in Memphis, Tennessee, provides a service of cotton testing to the cotton industry on more than one million bales per year. A 10-ounce sample is obtained from a bale of cotton and tested on instrumentation. The 10-ounce sample is discarded into an air duct that is contained beneath the floor in an enclosed trench. The sample is conveyed back to the bale press through an air removal system to a cotton condenser that separates the cotton from the air stream that drops through a sample buster into the bale press. The office can provide testing information on approximately 60,000 samples per day. The bale press capability should be 3,000 pounds per hour of loose cotton for the bale press to compress, tie and package. The press box shall be accessible for the operator

to tie the bale easily and safely. **The system must remove the bales from the press using an exit conveyor that is installed by the Contractor. The system must be able to transfer bale from exit conveyor to the bagging system that is installed by the Contractor. Once bales have been wrapped, the bagging system will eject the bale into an upright position on the heavy-duty bracing system that is installed by the Contractor.**

C.4 SPECIFICATIONS:

Bale Press Specifications:

- The bale press box dimensions must be 54" x 36" with a bale thickness of 24" which corresponds to the dimensions of a modified flat bale press at a minimum weight of 480 pounds. It shall utilize six 10'3" steel ties.
- The electrical requirements must be 460 volt 3-phase, which is currently provided.
- The controls must interlock with the existing conveyor system. The startup sequencing must be controlled in the following order.
 - Bale Press
 - Sample Buster
 - Conveyor
- The bale press must be automatically controlled by a PLC that determines the cycle of the ram, completed bale size (as determined by the hydraulic pressure, which shall also be adjustable) and the door opening and closure for the full ejection of the completed bale before the start of another bale.
- All exterior bare metal on both the bale press and conveyor system shall be powder-coated or painted with industrial enamel. It is preferable that both be the same color. Color must be approved in advance by the USDA Technical Point-of-Contact.
- An exit swing or pivot door at the exhaust of the bale that opens in a manner as to not obstruct the movement of the bale on the exiting conveyor rollers.

Steady Flow Breaker Specifications:

- The system must be capable of distributing cotton evenly throughout the bale chamber so the bales would consistently form.
- All exterior bare metal on steady flow breaker shall be powder-coated or painted with same industrial enamel as the baler and conveyance system.

Automated Conveyor System Specifications:

- Replace the current unload conveyor system with a more durable conveyor equipped with weighing capabilities.
- Entry point of automated conveyor must be at a height of 15" and a width of 64" with an incline to the height of 37".
- The exiting conveyor may include optional motorized power rollers.
- Conveyor Control Box featuring an independent Programmable Logic Controller (PLC). This control box must contain push buttons for the Emergency Stop (e-stop), manual/auto modes, start function, and an off/on switch, accompanied by appropriate indicator lights.

Bale Bagging System Specifications:

- The system and operation should correspond with the current bagging system USDA Cotton and Tobacco has in the Grading Department at the Memphis Classing Office.
- All exterior bare metal on the bale bagging system shall be powder-coated or painted with same industrial enamel as the baler and conveyor system.
- System must transfer bales from the automated conveyor at a height of 37”.
- Pictures and videos will be provided from USDA.

C.5 INSTALLATION TASKS:

The Contractor must perform the following tasks as a part of the installation of the new bale press, automated conveyor, bale bagging and heavy-duty bracing system:

- Disassemble and remove the existing bale press located at the C&T Program Cotton Classing Facility in Memphis, Tennessee.
- Install new full-eject bale press located at the C&T Program Cotton Classing Facility in Memphis, Tennessee.
 - Must include necessary railing to restrict access to hazardous areas (must meet all local and OSHA safety codes).
 - Must include an operator’s control panel that is labeled and easily understood for simple operation.
- Install steady flow breaker:
 - Must include necessary railing to restrict access to hazardous areas (must meet all local and OSHA safety codes).
 - Must be capable of distributing cotton evenly inside the baler chamber.
- Install automated conveyor system:
 - Include motorized power rollers.
 - An operator’s control panel that is labeled and easily understood for simple operation.
 - Location of conveyor system will be in the quality assurance baler room and air lock.
 - Must have weighing capabilities.
- Installation of a bale bagging system:
 - An operator’s control panel that is labeled and easily understood for simple operation.
 - Must include necessary railing to restrict access to hazardous areas (must meet all local and OSHA safety codes).
 - Must be capable of transferring bales from exiting conveyor into plastic bags efficiently.
 - Location of bagging system will be in quality assurance air lock.
- Installation of a heavy-duty brace system:
 - Must be able to protect walls and floor from damages.
 - Must be able to continuously withstand bales falling to ground in an upright position.
 - Include gravity rollers for bale to exit to ground.
 - Location of heavy-duty brace system will be in quality assurance air lock.
- If necessary, the Contractor must make modifications to the existing cotton transport chute to deliver samples into the newly installed bale press. All potential modifications

that interfere with the existing building structure or the catwalk must be approved by USDA engineering personnel.

C.6 TRAINING:

The contractor must provide start-up and training within two (2) weeks' notice by the USDA Technical Point-of-Contact by a manufacturer's representative and demonstrate proper bale press operation on a maximum of 10 bales or until USDA representatives on-site are satisfied. The USDA shall verify the operation procedures of the bale press and conveyor system, bale dimensions, and bale weight within a weight range of 485-515 pounds. The USDA shall provide all of the necessary loose cotton and labor to supply the cotton to the bale press. A punch list must be developed at this time which must be satisfied by the contractor before final acceptance.

C.7 WARRANTY:

The Contractor must provide a *minimum* **one (1) year** full-service warranty, including all parts, labor, supplies, etc., necessary to maintain the equipment in good operating condition at no additional cost to the USDA. The warranty period must begin after the installation of the new baler and upon the date the baler system (Maintenance service must not include electrical work external to the equipment or repair of damage resulting from accident, neglect, misuse, or causes other than ordinary use.